



OPEN Introducing an efficient method for feature extraction in image retrieval systems

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Image retrieval systems are of particular importance because they can be effectively used in image management in large databases as well as in search engines. By enabling fast and accurate image searches based on relevant features, these systems help users quickly access the desired information and significantly increase the efficiency of search processes. However, current systems usually do not cover the comprehensive features necessary to describe different types of images. In this research, we have tried to improve the performance of image retrieval systems in this aspect by using a wider set of features. Our proposed method generally consists of several steps. In the first step, we perform the feature description process, in which four feature categories are used to describe image characteristics. These categories include texture, color, and various patterns related to these two features. Then, a feature selection algorithm based on Black Hole Optimization (BHO) is used to extract an optimal subset of these features. This step helps us to improve the retrieval accuracy in addition to increasing the processing speed. Finally, a Fuzzy C-Means (FCM) is used to retrieve images based on the selected feature set. The performance of the proposed method was evaluated based on two datasets and achieved significant precisions of 0.97, 0.95, and 0.94 in the Corel1000, Brodatz, and ALOI databases.

Keywords Image retrieval, Feature extraction, Black hole optimization, Fuzzy C-Means, Gray level co-occurrence matrix

Increased digitization of all forms of information, from basic textual information to more complicated information like photographs or videos, is a reflection of the rapid advancement of technology that we are witnessing on a worldwide scale. At the same time, storage systems are being employed more and more, which leads to large databases for general use as well as specialized use in particular fields of expertise. Although there is a lot more information available now and it is quite easy to obtain, the growing size of databases makes it more difficult to locate the precise information one is looking for. In this regard, search engine efficacy and efficiency become crucial when used in specialized fields such as industrial property or medical diagnostics¹. A well-researched picture matching problem, image retrieval is a subfield of information retrieval² that has attracted significant scholarly attention since the 1970s. It involves recovering similar images from a database^{3,4}.

The fields of image processing and image storage technologies have seen significant advancement. All of this has contributed to the development of various kinds of enormous image repositories. However, managing these databases takes a lot of effort and time. Therefore, an effective and quick retrieval system is needed to search and index photos from these huge databases. Traditionally, images are retrieved using the information-Based Image Retrieval (TBIR) system, which requires human entry of textual descriptions, such as keywords or other information that describes the image⁵. Eventually, similar images are obtained from that specific term. However, this approach has a number of drawbacks, including homonyms, synonyms, and mistakes in human annotation. All of these drawbacks make TBIR an unreliable picture retrieval technique. The Content-Based Image Retrieval (CBIR) framework was created in order to get over the drawbacks of the TBIR system. It uses the visual characteristics of the images to automatically retrieve them without the need for human intervention⁶. While CBIR systems rely on the extraction of visual features, the main novelty of our approach arises from combining multiple visual features extraction methods with the optimization of these methods through a customized feature selection retrieval framework.

So, to construct this kind of system, the initial process is to extract low-level features of image and at the same time obtain the manner of image efficiently and productively and then, the similarity algorithm is used to search

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for the images. This paper focuses on three general recognized graphic characteristics, which are color, shape and texture⁷. These low level features are essentially inherent in the contents of an image. All the image annotation and retrieval systems have been built successfully employing these features. The results of the recall, precision, and fused measures substantially improve when all the three aspects are included. But individually, color and texture are the most important and frequently used features since their extraction is easier than shape features and is somehow more accurate⁸.

The research gaps in the field of image retrieval focus on the diversity of features that current systems cover. Current systems usually use only a limited range of features to describe the content of images. This limitation prevents us from providing optimal performance for applications that require a general and generalizable image retrieval model. In particular, image retrieval models in previous research often have to be designed and tuned to the characteristics of the dataset used or the type of data (images to be retrieved). In this regard, our proposed method aims to expand the set of features that are used and, critically, to apply an optimization technique (Black Hole Optimization-BHO) to dynamically select an optimal and non-redundant subset of these features, in order to provide adaptive and efficient retrieval models. Finally, the optimized feature set is used as an input to a FCM (Fuzzy C-Means) clustering algorithm to build a novel retrieval model. Therefore, our study aims at addressing the gaps in the presentation and the better utilisation of features in image retrieval systems.

Our research innovation is divided into two main parts. First, we select the image features of CNN (Convolutional Neural Network), GLCM (Gray Level Co-occurrence Matrix), HA (Histogram Analysis), and statistical features of images to describe the characteristics of images. This combination enables us to take the best of each technique and improve the performance and accuracy of the image retrieval system. The second innovation of our research is the application of the BHO algorithm for feature selection and formation of an adaptive image retrieval model. This algorithm enables us to rein out inefficient features or, in other words, to choose only effective features to enhance the retrieval. The following describes the contributions of this paper in feature extraction for image retrieval systems:

- Using the BHO algorithm to select the optimal feature from a diverse set of features extracted via CNN, HA, and GLCM helps to reduce processing time and increase retrieval accuracy by removing irrelevant features.
- Employing a set of image analysis techniques, including CNN, HA, and GLCM, enables the extraction of rich and diverse features from images. The CNN was configured to extract a compact and effective feature vector that is well integrated with the other feature extraction methods.
- Using a fuzzy clustering algorithm to classify the BHO-optimized features facilitates the development of an effective model for retrieving new images and performing more accurate queries.

The remainder of the document is structured as follows: Related research is reviewed in Sect. 2. The suggested method is thoroughly explained in Sect. 3. The findings of the implementation are discussed in Sect. 4, and conclusions on the findings of the suggested strategy are described in Sect. 5.

Related work

This section is dedicated to reviewing some of the recent attempts for CBIR. Vo et al.⁹ investigated image retrieval utilizing a query and text for specific alterations, introducing a novel method that surpassed existing techniques across three datasets and executed image classification with compositionally innovative labels on MIT-States. Hosseinzadeh et al.¹⁰ proposed a new approach to solve composed query image retrieval using local areas of the image and set up the relationship between modification text and image regions. Experimental results on three benchmark datasets proved that the approach outperformed other approaches.

Liu et al.¹¹ proposed a bi-directional training scheme for composed image retrieval, which better enhances the model performance with reversed queries and fine-tunes the parameters of the text embedding module without much modification of existing architectures. Liu et al.¹² expanded the Compose Image Retrieval on Real-Life Images (CIRR) dataset to encompass intricate images, presenting CIRPLANT, a transformer-based model that employed pre-trained vision and language expertise to adjust visual attributes according to natural language parameters.

Jandial et al.¹³ proposed a novel framework known as Semantic Attention Composition (SAC) for text-conditioned image retrieval and showed superiority over the existing ones on benchmark datasets like FashionIQ, Shoes, and Birds-to-Words, while naturally supporting natural language feedback. Tian et al.¹⁴ presented Additive Attention Compositional Learning (AACL) for fashion picture retrieval utilizing textual feedback, showcasing its applicability in e-commerce using a multi-modal transformer-based architecture.

Baldrati et al.¹⁵ proposed a state-of-the-art image retrieval approach for e-commerce applications using Contrastive Language-Image Pre-training (CLIP) features and contrastive learning for fine-tuning the text encoder. Devulapalli et al.¹⁶ proposed a hybrid feature extraction technique for CBIR using both high- and low-level features that attained a precision of 91%, outperforming existing algorithms for classification and object detection.

Wangi and Makandar¹⁷ employed Autoencoder, a deep learning neural network methodology, to decrease image retrieval duration and enhance outcomes using CBIR. They utilized Simple, Deep, and Convolutional autoencoders. Keisham and Neelima¹⁸ developed a CBIR system utilizing a deep Search and Rescue (SAR) algorithm to improve image retrieval efficiency, enhancing precision and recall using pre-processing, feature extraction, fusion, clustering, and classification, which was evaluated on four databases.

Shamsipour et al.¹⁹ proposed a hybrid feature set, which combined the power of classical feature engineering techniques with deep CNN for image retrieval, recording accuracies of 96.68% and 94.56% on benchmark datasets, thus giving promising results compared to the state-of-the-art machine learning and deep learning-based techniques. Chhabra et al.²⁰ introduced a CBIR system that employed Oriented FAST and Rotated BRIEF

in conjunction with scale-invariant feature transform to extract feature vectors from digital images, evaluating its efficacy through decision trees and random forests.

Nesakumari²¹ presented three CBIR techniques: Surrounding Information Retrieval (SIR), Minimum Edge Retrieval (MER), and Integrated Feature Retrieval (IFR), which collected features from adjacent intensity values to enhance precision. Sikandar et al.²² proposed a hybrid deep learning- and machine learning-based CBIR system that efficiently quantified similarity between database and query images, outperforming other systems and proving suitable for various applications. Salih and Abdulla²³ proposed a hybrid CBIR technique that employed two layers of filtering to compare query images with database images based on texture and color features.

Rani et al.²⁴ used Content-Based Medical Image Retrieval to derive significant information from Computed Tomography scans, attaining precision, recall, F-measure, and accuracy with fractional Hartley transform and hybrid feature extraction. Taheri et al.²⁵ presented a deep feature fusion technique for image retrieval utilizing a segmented neutrosophic set, enhancing outcomes and minimizing semantic discrepancies by analyzing images into true, false, and indeterminate levels. Sudhish et al.²⁶ introduced content-based medical image retrieval pipeline employing CNN models and clustering techniques, attaining great accuracy and displaying practical relevance in brain tumor diagnosis via Magnetic Resonance Imaging.

Mahji et al.²⁷ focused on the privacy preserving aspect of CBIR. Their approach employed k-means clustering with silhouette index for constructing the CBIR model, and random projection for encryption of image features. This model also devised locally likely arrangement hashing technique to match the encrypted features. This approach is appropriate for application such as medical image retrieval where privacy is a key factor. Yenigalla et al.²⁸ introduced a CBIR system based on neural networks. In this model, low-level and high-level features are extracted from images. The former includes a set of shape, texture, and color attributes; while the latter includes a series of features extracted through machine learning algorithms. The extracted features are classified using a CNN for retrieval purposes. Sezavar et al.²⁹, suggested a combination of sparse representation and CNN for CBIR. In their approach, a CNN consisting 5 convolution layers is employed for extracting deep-features from the images. After obtaining query feature vector, the sparse representation mechanism is used for retrieving relevant images which is shown to be effective in raising speed and accuracy.

Research methodology

In this section, after describing the characteristics of the data set used, the details of the proposed model will be presented.

Database

To evaluate the proposed method, three different databases have been employed. The first database is Brodatz³⁰. This database consists of a collection of images with various textures that have been rotated at different angles. It consists of 91 images having 13 different texture classes. The images in this database have been rotated in the manner shown below 0, 30, 60, 90, 120, 150, 180. All the images in this database have a size of 512 by 512 pixels. Some of the images in this database are shown in the Fig. 1-a below. The second database used is the Corel1000 database³¹. This is a database of color images that has been used to assess color image search techniques in different researches. It comprises of 1000 images that have been grouped into 10 classes. The size of each picture in this database was changed to 384 by 256 pixels. Some images from this database are illustrated in the Fig. 1-b. The third dataset is Amsterdam Library of Object Images (ALOI)³². This dataset includes 1000 categories, each of which includes a series of images belonging to an object. We employed the full color (24 bits) half resolution (384 × 288 pixels) version of this dataset which includes 72 variations of viewing direction for each object category. Some images from this database are illustrated in the Fig. 1-c.

Proposed model

In the following sub-sections, the authors will describe in detail the implementation of the proposed approach for color image retrieval based on the utilization of deep learning in conjunction with optimization and clustering. The method under development involves the use of CNN, HA and GLCM for feature extraction of images from the database. The BHO algorithm is then applied to the images of the database to choose the best subset of the image features that has been extracted. The intended use of this algorithm is to obtain the maximum precision in identifying the right images and at the same time minimizing the time it takes for this algorithm to complete. Afterwards, feature selection is applied and a fuzzy clustering algorithm is used for image classification and the model construction. The proposed method finds the images similar to the query image using the fuzzy distance measure. The block diagram illustrating the proposed method is shown in the Fig. 2. We will offer more information on each of the sections. As demonstrated in Fig. 2, the presented CBIR mechanism is defined by the following processes:

- **Feature Extraction:** In this step, the features of the input image are extracted using a set of CNN models, HA, and GLCM, along with features that describe the color correlation of pixels.
- **Feature Selection:** In this stage, an optimal subset of the generated features is selected by the utilization of BHO algorithm. This is to enhance the efficiency of the proposed method by reducing the number of features on which the retrieval is based while at the same time enhancing the accuracy of the retrieval.
- **Feature Clustering:** Here, the features chosen by the BHO algorithm are classified using a fuzzy clustering algorithm. This clustering is used as a model for querying new images and for performing the retrieval operation.

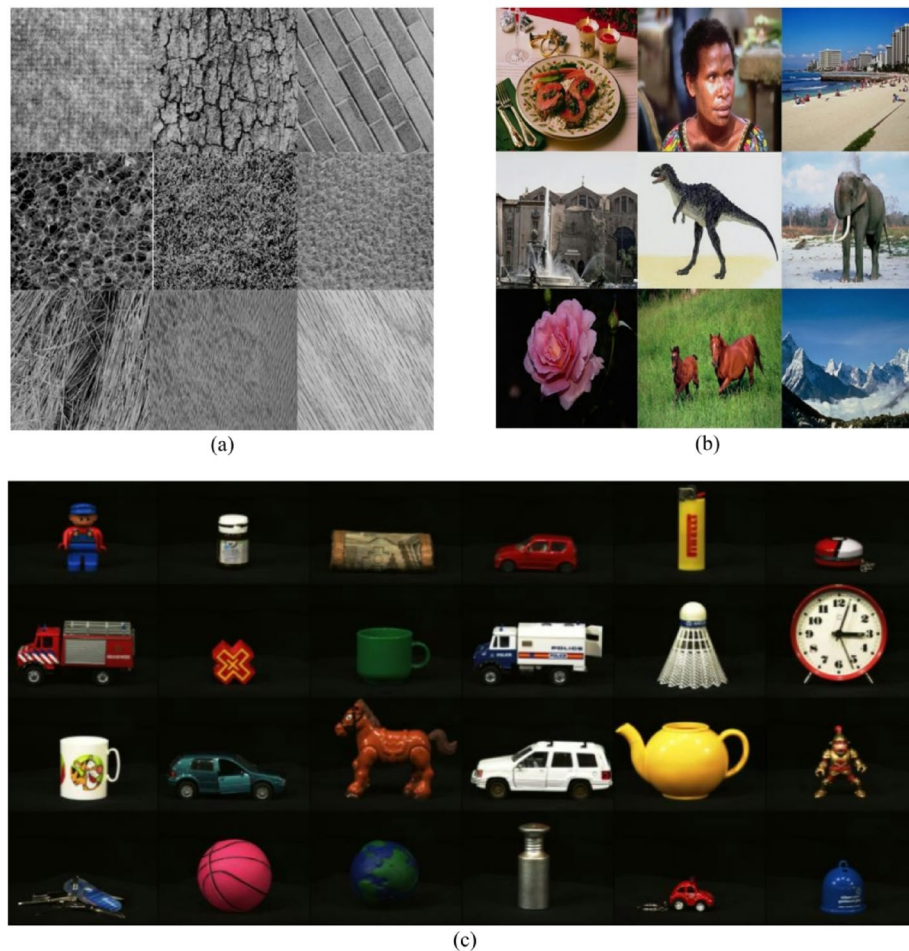


Fig. 1. Examples of database images (a) Brodatz³⁰, (b) Corel1000³¹, and (c) ALOI³².

Image feature extraction

The proposed method is aimed at processing and search of color images. This makes it possible to obtain and use the features that depend on the color of images. Accordingly, several feature extraction methods are employed in the proposed approach for processing color images:

- HA.
- Color Correlation of Pixels and Color Attributes of the Image.
- GLCM.
- CNN.

a) feature extraction based on HA.

The HA is the first method employed for extracting features from color images. In this algorithm, the color system of the image is first converted to the HSV format. Then, the frequency of color values for each layer H, S, and V will be analyzed separately. To achieve this, the input image I with dimensions $W \times H$ is mapped to the space $8 \times 2 \times 2$. Specifically, for the H layer, eight different levels are used for counting the frequency of color values, while two levels are considered for the S and V layers. To map the pixel values of each layer to the new space, each pixel's value is first quantized using the following equation:

$$q_{i,j} = \lceil N \times \frac{p_{i,j}}{\max_p} \rceil \quad (1)$$

In the above equation, N represents the number of levels for the current layer, $p_{i,j}$ is the pixel value at row i and column j in the current layer, and $\max_p$ is the maximum value corresponding to this layer. After quantizing the values of each pixel, a matrix Q will be obtained for each color layer to count the frequencies of the quantized values. To generate the HSV histogram of the image, a $8 \times 2 \times 2$ matrix named H is considered, and for each pixel in the image, the frequency in all three layers is counted. For example, if a pixel in the first layer has a value of 7, in the second layer 1, and in the third layer 2, we increment the value of H at row 7, column 1, and layer 2 by

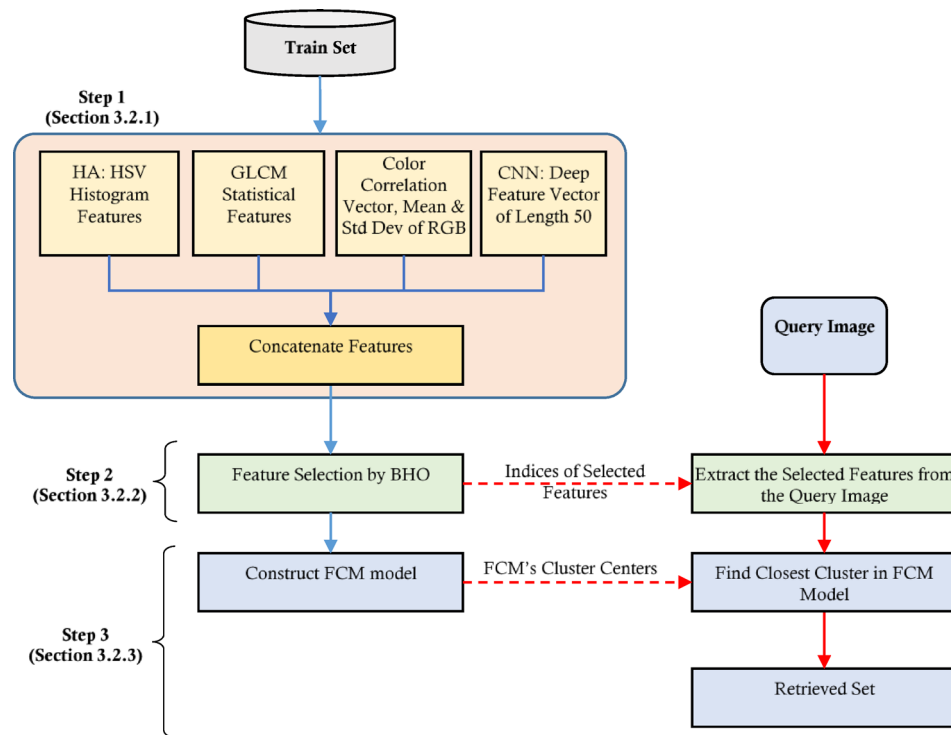


Fig. 2. Block Diagram of the Proposed Method.

one unit. After obtaining the matrix H , it is converted into a vector and considered as the HSV histogram of the image.

b) *Feature extraction based on pixel color correlation and image color Attributes.*

The second image feature extraction approach is based on color correlation between pixels. For this purpose, the input image I in the RGB color system is mapped first to 64 colors in the RGB space with dimensions $4 \times 4 \times 4$. This is achieved through a quantization process repeated for every pixel by using Eq. 1. The resulting quantized matrix is considered the image I' with a limited color space.

In the image I' , every color layer has four levels, thus, enabling each pixel to take one of 64 colors available in the mapped space. A radius is assigned to the neighborhood of the pixels in the image. In case of any given pixel, it firstly identifies the number of pixels within the defined radius, then compares the current pixel against its neighbors. If the current pixel value happens to match the neighbor value, then that pixel color's frequency count increases by one unit. The process would yield a vector of length 64, which is representative of the color correlation of the pixels in the image based on the neighborhood radius. Values are normalized on this vector in order to map them to the range $[0, 1]$. This operation is done for 4 neighborhood radii with the proposed methodology and it stores data from the results of a pixel color correlation as a feature vector.

The color correlation features of the pixels perform the image description at the pixel level. In the proposed method, features at the layer level are also used to describe the database images. We refer to this set of features as the color attributes of the layers. To extract this category of features, we decompose the input color image into its constituent layers. Following that, for each layer, the mean and standard deviation of the pixel brightness in that layer are calculated. By doing this for each layer, we describe the feature vector representing the color attributes of the image as $\langle M_R, S_R, M_G, S_G, M_B, S_B \rangle$. In this set, M_R denotes the mean brightness of the red layer, and S_R represents the standard deviation associated with the red layer. Additionally, the indices B and G correspond to the attributes of the blue and green layers, respectively.

c) *Feature Extraction Based on GLCM.*

The third category of features used in the description of an image is that of GLCM. GLCM refers to the probabilistic model for describing the distribution of the pixel brightness values in adjacent pixels within an image. It is a two-dimensional probability function considering the brightness of pixel i at one position and the brightness of pixel j at another position. Features that are usually extracted from GLCM are in the form of a statistical measure which describes image characteristics. Such features can be used for describing various patterns in images. Some of the GLCM features that could be used in describing images include:

Correlation Degree: The feature measures the degree of correlation between the brightness values of neighboring pixels. A high value of correlation will indicate that the immediate neighbors all have about the same brightness. This feature may be used to locate areas that contain regular patterns.

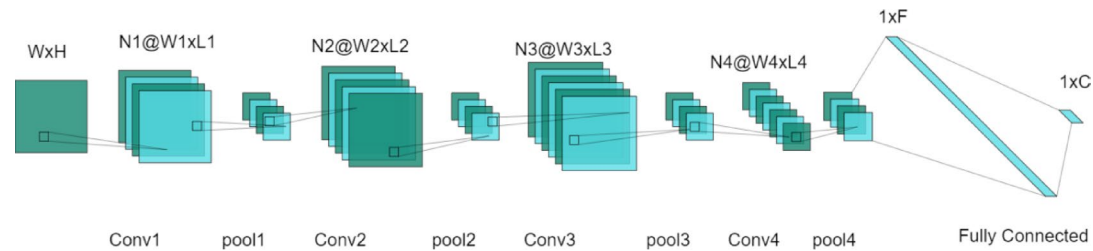


Fig. 3. Structure of the proposed CNN model for extracting image features.

Hyperparameter	Search Values
Convolution Width/Length	{2, 3, 4, 6, 7}
Number of Filters	{8, 16, 24, 32, 48, 64}
Activation	ReLU, PReLU, LeakyReLU
Pooling	Max, Min, Average
Optimizer	SGDM, Adam
Mini Batch	{8, 16, 32}

Table 1. Hyperparameters considered for configuring the CNN feature extraction Model.

Variance: This feature provides information about the dispersion of the data. A high value of variance indicates that the data values are highly dispersed. This feature can be utilized for detecting those regions having a high variance.

Flatness Coefficient: It measures the uniformity of the brightness distribution. The high value of the flatness coefficient means brightness distribution is uniform. It can be used to locate the areas where brightness distribution is uniform.

The features above are then extracted as the third category of image descriptors to work with other categories of features as input for the proposed retrieval model.

d) *feature extraction based on CNN.*

The last category of features describing the images are extracted through a CNN model. Figure 3 depicts the CNN structure utilized in the suggested technique for picture feature extraction. An attempt has been made to adopt the most straightforward framework for feature extraction in the architecture of this CNN model. Following an analysis of various CNN model designs for feature extraction, it was found that a model with three layers of convolution, three pooling layers, and two fully connected layers would perform the best.

Figure 3 shows that the result of the last fully connected layer of the CNN is regarded as the features that were extracted for every instance, and that it lacks the layers required for feature categorization. The matrices of the input color images are resized to dimensions 100 × 100 to be applied as input to the CNN model at the input layer.

The extraction of feature maps from the input patterns is carried out by three convolutional layers comprising 8, 24, and 48 filters in that order. The sizes of these three filters will be 6, 5, and 3, respectively. Further, the extracted feature maps have been subjected to a max pooling layer of size 2 × 2 for reducing the dimensions of feature maps. It is important to note that the stride for the convolutional and pooling layers is 1. After each of the pooling layers, an ReLU layer follows for the activation. The collected patterns are then transformed into vector form using two successive completely linked layers, which also reduce the feature matrix's dimensions to 50. This CNN model's feature vector is then used as an input for the suggested retrieval model.

In order to achieve a model with the highest efficiency in feature extraction, a comprehensive search strategy has been used to adjust the hyperparameters of the CNN model. During the search phase, hyperparameters pertaining to the CNN configuration are analyzed, including the convolution layer's size, the amount of filters, function of activation type, and pooling function kind. Additionally, the model's performance has been examined for various training setups, including minimum batch size and training procedure. The hyperparameters taken into consideration when configuring the CNN model are provided in Table 1.

The validation error criterion has served as the basis for the configuration scan and its suitability assessment. Consequently, according to the thorough search, the CNN model configuration (Fig. 3) detailed in this part has the smallest validation error. The greatest performance in the used data set, however, was obtained via CNN model training using the Adam optimizer and a minimum batch size of 16.

Feature selection using the BHO algorithm

After extracting the features related to each image, the feature selection process is carried out using the BHO algorithm. The goal of this algorithm is to reduce the dimensionality of the features in order to increase processing speed and decrease the error rate in image retrieval. In this section, we will detail the fitting function

of the algorithm to extract the best features. To achieve this, the issue of selecting prominent features needs to be modeled as an optimization problem.

Each candidate feature's selection status is regarded as an optimization variable in the optimization model utilized in the suggested approach. Because each optimization element in the BHO method can have a value of either zero (not picking the feature) or one (selecting it), the total amount of optimization elements in the algorithm is equal to the total amount of candidate features. In the BHO approach, each answer is thus represented as a binary vector having the length of N , wherein N is the collection of characteristics that were retrieved in the preceding phase. Each index vector can represent a possible response for how to select the prominent features in describing the image information.

Correlation criteria are used to choose features in the suggested optimization model. In this instance, the fewest features that are most closely associated with the image's semantic category are taken into account. Furthermore, redundant information shouldn't be present in the feature set that was chosen. This condition is met by calculating the correlation between the chosen features and attempting to reduce this value. As a result, the following formula is used to determine how well each answer vector fits the suggested optimization model:

$$Fitness(\vec{x}) = \frac{\frac{1}{K^2} \sum_{i=1}^K \sum_{j=1}^K |Corr(\vec{x}_i, \vec{x}_j)|}{\sqrt{K} \left(1 + \frac{1}{K} \sum_{k=1}^K |Corr(\vec{x}_k, Q)|\right)} \quad (2)$$

The total amount of features chosen in a solution is indicated by K in this equation, and they can be obtained by adding up all of the solution vector's values of one. The value of the i -th chosen characteristic in the solution $\vec{x}$ is described by $\vec{x}_i$. Lastly, Q indicates the quantitative vector of the image's conceptual category, which is regarded as the goal variable. The function $Corr(X, Y)$ shows the correlation across the values of two attributes, X and Y . The aforementioned equation's numerator looks to find characteristics that have the least amount of correlation with one another, while the denominator attempts to reduce the total number of features chosen in addition to assessing how well the chosen features correlate with the goal variable. The BHO method uses the following processes to identify salient features that best describe the image's semantic information based on the suggested model:

- Step 1: Create a starting population of binary strings representing solution vectors at random.
- Step 2: Using Eq. (2), determine each solution vector's fitness.
- Step 3: Determine which solution is the black hole with the lowest fitness.
- Step 4: Adjust each solution's location as indicated below³³:

$$X_i = X_i + rand. (X_{BH} - X_i) \quad (3)$$

In (3), the vector X_{BH} represents the position of the black hole, while the vector X_i denotes the position of the star/solution i in the solution space. Additionally, $rand$ is a random number in the range (0, 1).

- Step 5: Calculate the threshold distance for the star to be swallowed by the black hole as follows³³:

$$R = \frac{fitness(BH)}{\sum_{i=1}^N fitness(i)} \quad (4)$$

In this equation, N represents the number of solution vectors in the BHO algorithm.

- Step 6: If the fitness of a solution vector is lower than that of the black hole, replace the current black hole with it.

- Step 7: Remove the solution vectors that are within the threshold distance R from the black hole and replace them with a new randomly generated solution vector.

- Step 8: If the number of iterations of the algorithm has reached the threshold T , proceed to the next step; otherwise, repeat the search from Step 2.

- Step 9: Return the black hole with the lowest fitness as the best discovered solution.

After determining the optimal response based on the above steps, the selected features are extracted from this response, forming the input for the proposed clustering model in Step 3.

Building the retrieval model using FCM clustering in the proposed method

In the proposed method, the FCM clustering model is used to construct the image retrieval model based on the BHO-optimized feature set. During the experiments and the evaluation of various clustering algorithms, it was found that this algorithm exhibits the highest efficiency in image retrieval. Therefore, the FCM algorithm has been employed to build the retrieval model. To construct the retrieval model, the features selected by the BHO algorithm are provided to the FCM clustering algorithm to categorize the database images. In the proposed method, the number of clusters is set equal to the number of image categories in the database. After clustering the features, the centers of the clusters are identified. One of the most important and widely used clustering algorithms is the FCM algorithm. In this algorithm, samples are divided into c clusters, with the number of clusters c predetermined. The steps for clustering the data using the FCM algorithm include the following:

- Initial cluster approximation as follows³⁴:

$$u_{ik} = \frac{1}{\sum_{j=1}^c \left(\frac{d_{ik}}{d_{ij}}\right)^2} \quad (5)$$

In the above equation, u_{ik} represents the degree of membership of sample i in cluster k . Additionally, d_{ik} indicates the difference between sample i and cluster k .

Calculate the centers of the clusters as follows³⁴:

$$v_k = \frac{\sum_{i=1}^n u_{ik}^2 x_i}{\sum_{i=1}^n u_{ik}} \quad (6)$$

In the above equation, v_k represents the center of region k , and x_i denotes the i -th feature of the image. Additionally, n indicates the total number of features of the input image.

If $|U_{i+1} - U_i| \leq \epsilon$, the algorithm terminates; otherwise, the aforementioned steps are repeated.

In the steps outlined in this section, the clustering is performed in an unsupervised manner. It is important to note that if the output classes are known and the goal of clustering the samples is based on these output classes, the clustering will take place in two stages: model construction and testing. In the first stage, an FCM model consisting of the centers of the fuzzy clusters will be formed. This model is created through steps 1 to 3 described above, with the difference being that the vector u_{ik} in the first step will correspond to the classification of the samples, and consequently, the centers of the clusters will be determined based on the target classes. Then, in the testing phase, the established FCM model will be used for clustering the test samples. In the proposed method, to retrieve images related to a queried image, the selected features of the input image are first collected, and the distance between the input image and the center of each cluster in the retrieval model is calculated. The queried image is categorized into the cluster whose center is closest to it. For image retrieval, K images in the same cluster that have the smallest distance to the queried image will be returned as the retrieval results. Pseudo-code of the proposed image retrieval system is presented in Algorithm 1.

Algorithm: Proposed Image Retrieval System**Input:**

Image Database: Set of all images; $Query_{Image}$: Image to be retrieved; C: Number of clusters (FCM); T: Max iterations (BHO); epsilon: FCM stopping criterion

Output:

Retrieved Images: K most similar images to Query Image

// 1. Feature Extraction

Function: ExtractFeatures(Image)

// 1.1 HA Feature Extraction

HSV_{Image} = Convert Image to HSV color space

$HA_{Features}$ = Calculate HSV Histogram of HSV_{Image} using Equation (1)

// 1.2 Color Correlation & Attributes Feature Extraction

$Quantized_{Image}$ = Quantize Image to 64 colors using Equation (1)

$ColorCorrelation_{Features}$ = Calculate color correlation of $Quantized_{Image}$ for different radii

$ColorAttribute_{Features}$ = Calculate mean and standard deviation of RGB layers

// 1.3 GLCM Feature Extraction

$GLCM_{Features}$ = Calculate GLCM features (Correlation, Variance, Flatness)

// 1.4 CNN Feature Extraction

$Resized_{Image}$ = Resize Image to 100×100

$CNN_{Features}$ = Extract features using CNN model (Figure 3)

$Combined_{Features}$ = Concatenate ($HA_{Features}$, $ColorCorrelation_{Features}$, $ColorAttribute_{Features}$, $GLCM_{Features}$, $CNN_{Features}$)

Return $Combined_{Features}$

// 2. Feature Selection using BHO

Function: SelectFeaturesBHO (Features, T)

Initialize Population: Generate random binary vectors (solutions)

For each solution in Population:

Calculate Fitness using Equation (2)

BlackHole = Solution with minimum Fitness

iteration = 0

While iteration < T:

For each solution in Population:

solution = solution + rand() * (BlackHole – solution) // Equation (3)

Calculate Fitness of solution using Equation (2)

If Fitness(solution) < Fitness(BlackHole):

BlackHole = solution

R = Fitness(BlackHole) / Sum of Fitness of all solutions // Equation (4)

For each solution in Population:

If Fitness(solution) is within R from BlackHole:

Replace solution with a new random solution

iteration = iteration + 1

$Selected_{Features}$ = Features corresponding to 1s in BlackHole

Return $Selected_{Features}$

// 3. Building Retrieval Model using FCM Clustering

Function: BuildRetrievalModelFCM($Selected_{Features_{Database}}$, C, epsilon)

// Initialize membership matrix U randomly

U = Random initialization of membership matrix

iteration = 0

While True:

$Cluster_{Centers}$ = Calculate Cluster Centers using Equation (6)

U_{new} = Calculate new membership matrix using Equation (5)

If $|U_{new} - U| < \text{epsilon}$:

Return $Cluster_{Centers}$

$U = U_{new}$

iteration = iteration + 1

Algorithm 1: Pseudo-code of the proposed image retrieval system

```

// Main Algorithm
// Feature extraction for each image in database
For each Image in Database:
    ImageFeatures = ExtractFeatures (Image)
    AllFeatures.Append(ImageFeatures)
// Feature Selection for Database Images
SelectedFeaturesDatabase = SelectFeaturesBHO (AllFeatures, T)
// Build retrieval model using FCM
RetrievalModel = BuildRetrievalModelFCM (SelectedFeaturesDatabase, C, epsilon)
// Feature extraction of query image
QueryFeatures = ExtractFeatures (QueryImage)
// Feature Selection for Query image
SelectedQueryFeatures = Filter the BHO-selected features in QueryFeatures
// Find the closest cluster
ClosestCluster = Find cluster in RetrievalModel closest to SelectedQueryFeatures
// Retrieve K nearest images within the closest cluster
RetrievedImages = Retrieve K images in ClosestCluster with minimum distance to SelectedQueryFeatures
Return RetrievedImages

```

Algorithm 1: continued

Research finding

The proposed method has been independently evaluated using three datasets Corel1000, Brodatz, and ALOI. To evaluate this method, a cross-validation technique with 10 folds was used and its implementation was done using MATLAB software version 2020a.

Evaluation metrics

In this research, the Precision, Recall, F-Measure and Area Under the Curve (AUC) metrics were used as tools to evaluate the model performance. Specifically, the evaluation operation was performed based on the retrieval process so that in each iteration, 10% of the database images are considered as unseen images. The retrieval model was developed using the remaining 90% of the data. Then, to each unseen image, the desired image is fed to the model and the retrieval operation is carried out with respect to it. This process was repeated 10 times and each iteration, a new set of images was considered as unseen instances. Therefore, after 10 iterations all of the instances were employed as query images. In each retrieval process, three images are taken out; Thus, three images are retrieved in Corel1000 database and similar number of images are taken out for Brodatz and ALOI databases. At last, these retrieved images are matched with the actual batch of the query image to find out the values of the precision, recall and F-Measure. This approach to the evaluation enables us to assess the effectiveness of the model in terms of image retrieval.

In Eqs. (7)–(9), True Positive, False Positive, False Negative, Precision, Recall, F-Measure and AUC represent various evaluation metrics used to assess the performance of the model.

- True Positive: The model correctly identifies related images as relevant.
- False Positive: The model incorrectly identifies unrelated images as relevant.
- False Negative: The model incorrectly identifies related images as irrelevant.

Precision: It is a measure about the model identifying positive samples correctly out of the total actual positive instances.

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad (7)$$

Recall: It defines the performance of the model in detecting the positive samples from an actual pool of positives. It is also defined as the sensitivity or the true positive rate.

$$Recall = \frac{True\ Positive}{False\ Negative + True\ Positive} \quad (8)$$

F-Measure: The harmonic average of the precision and recall yields the measure of F-Measure.

$$F - Measure = \frac{2 \times Precision \times recall}{Precision + recall} \quad (9)$$

The AUC: Its value is used to describe the predictive capabilities of the model, which ranges from 0 to 1, and values close to 1 indicate high model ability.

In addition, we performed the experiments in four different cases. The first case is the proposed method, in which the entire structure described in Sect. 3 is used. The second case “Prop. (CNN)” refers to the case where we only use the features extracted by CNN and perform other operations based on it. The third case “Prop. (GLCM)” only uses the features extracted using GLCM. Finally, the fourth case “Prop. (All Features)” refers to the case where the feature selection operation is not performed using BHO and all the features extracted from the four sets are used. We also used the models presented in ORBSIFT²⁰, Tamilkodi et al.²¹ Majhi et al.²⁷, Yenigalla et al.²⁸, and Sezavar et al.²⁹ for comparison.

Results

This section reports and discusses the results of the experiments based on the metrics mentioned in Sect. 4.1. Figure 4 shows the precision metrics on the Corel1000, Brodatz, and ALOI datasets. The left figures contain the precision-per-iterations plots for both available datasets. The proposed method in figures (a), (c) and (c) refers to the full use of all techniques and their combination, which includes CNN, HA, and GLCM feature extraction. This combination allows the model to exploit the advantages of all three techniques and thus provide high precision in image retrieval. Thus, in the Prop. (CNN) method, only the features extracted via CNN are used and a relatively good precision of about 0.88, 0.76, and 0.86 is achieved for Corel1000, Brodatz, and ALOI. In the third case, Prop. (GLCM), only the GLCM features are considered, and the precision is about 0.80, 0.84, and 0.78 which shows the limitations of this technique alone compared to the combination of several methods. Also, compared to the method of Tamilkodi et al.²¹ as the closest approach, the proposed method achieved an average superiority of 8.18%, 10.34%, and 4.16% for Corel1000, Brodatz, and ALOI datasets, respectively.

The figures on the right show the box plot of the precision variations. In general, this box plot consists of four main parts, each part representing the values of the precision changes in the first to fourth quartiles of the data distribution. This box plot provides useful information about the distribution and dispersion of the data and visually indicates how the data is distributed in these quartiles. The red line in the middle of the plot shows the

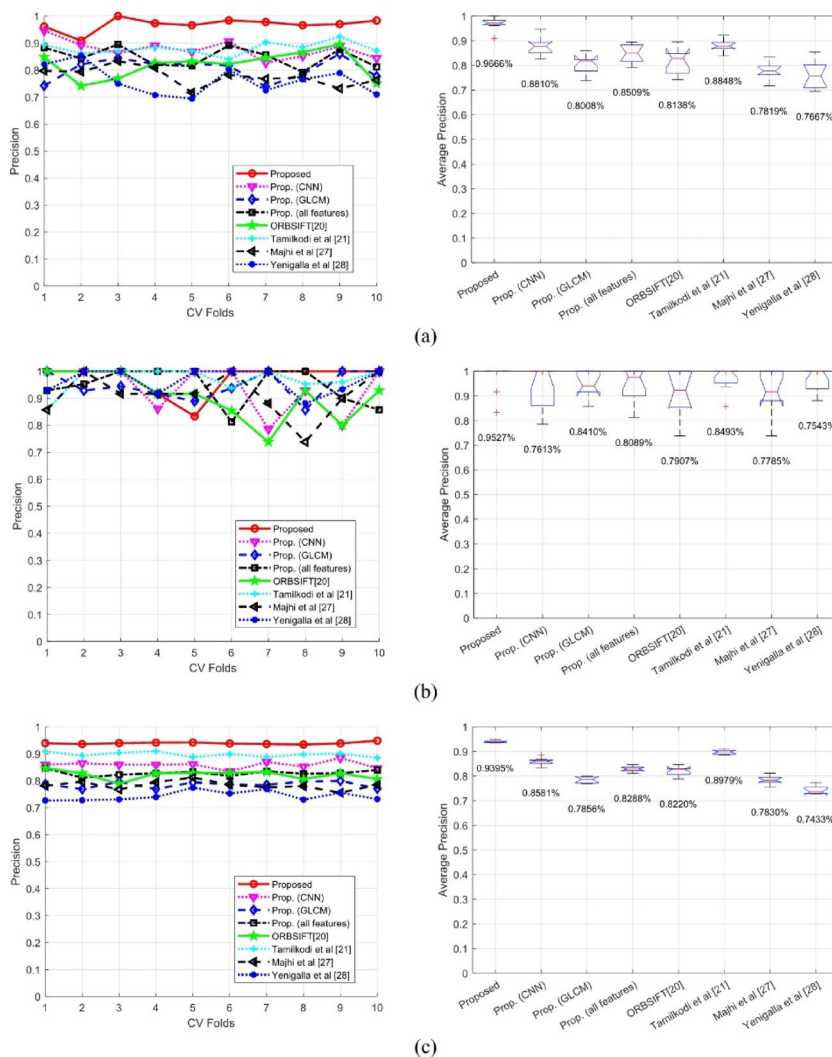


Fig. 4. Evaluation of precision metric and average precision (a) Corel1000, (b) Brodatz, and (c) ALOI dataset.

median precision and allows us to easily identify the center point of the precision distribution; in this case, the proposed method achieved an average precision of 0.953 for all datasets. Interestingly, in the proposed model, the median precision and the average precision for all datasets are very close, which indicates the success of this method in experiments with a higher level of confidence. This agreement indicates the high stability and reliability of the proposed method, as its results remain consistently high and close to the median level under different conditions. Therefore, it can be concluded that the use of the proposed hybrid method not only led to an increase in image retrieval precision, but also showed higher flexibility and efficiency than the comparative methods. Figure 5 shows the Recall metrics on the Corel1000, Brodatz, and ALOI datasets. The left figures show the recall-per-iteration values of both datasets at 10 folds.

In this graph, the proposed method in Fig. 5(a) has clearly succeeded in achieving recall increases of 0.05 and 0.04 over Prop. (GLCM) and Prop. (All Features) methods, respectively. This indicates a significant improvement in the model's ability to identify positive cases in the dataset. Also, in Fig. 5(b), the recall values have increased by 0.06 and 0.09 over similar methods, respectively. The results in Fig. 5(c) show improvements of 0.08 and 0.06 for the same comparisons. These results emphasize that the proposed method, by combining different techniques, has been able to provide better performance than other methods. By observing the recall results in the box plot on the right, the performance of the proposed method can be evaluated in comparison with other methods (such as Tamilkodi et al.²¹). Specifically, the recall values for this method are 0.2756, 0.4673, and 0.5042 for Corel1000, Brodatz, and ALOI datasets, respectively. The higher the boxplot of the proposed method is compared to the other methods, the better the accuracy in identifying true positives. Furthermore, the proximity of the median line to the highest boxplot values indicates the high stability and reliability of the proposed method under different conditions.

Figure 6 shows the F-Measure metrics on the Corel1000, Brodatz, and ALOI datasets. The left figures present the F-measure for both datasets at each of the 10 folds in the employed cross-validation approach. F-measure can be used as a very important and effective tool in evaluating the methods. The proposed method has

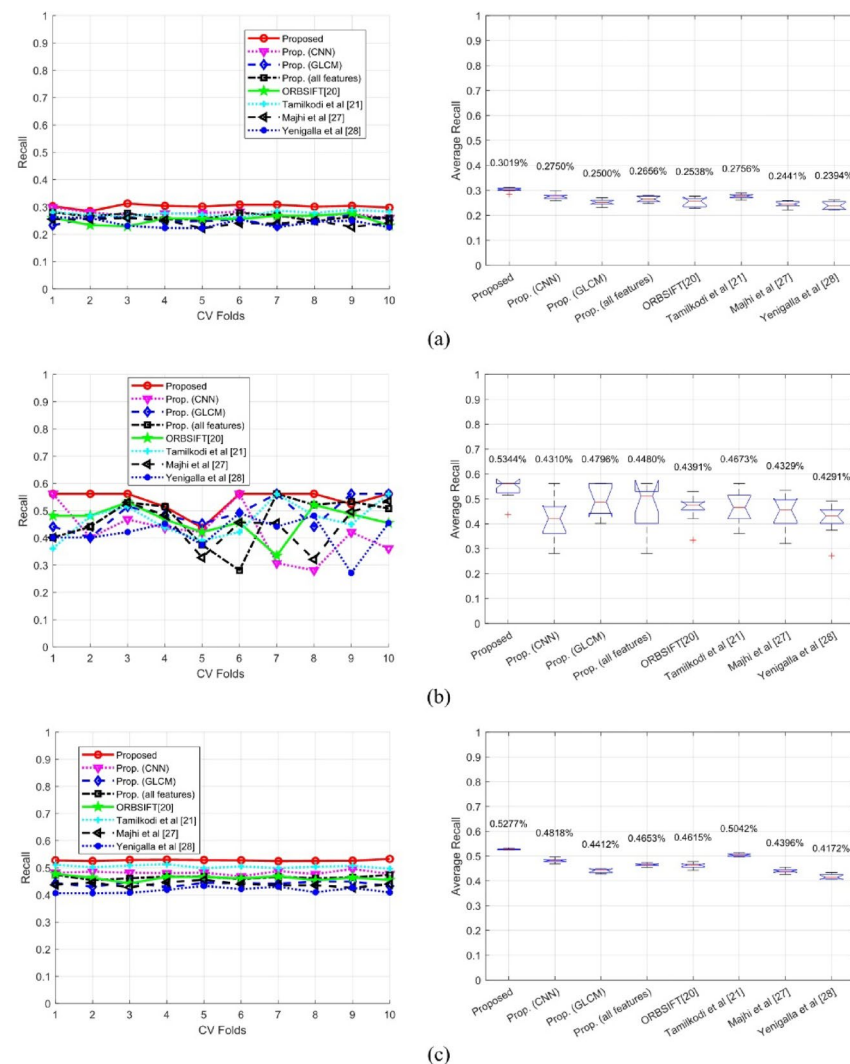


Fig. 5. Evaluation of Recall metric and average Recall (a) Corel1000, (b) Brodatz, and (c) ALOI dataset.

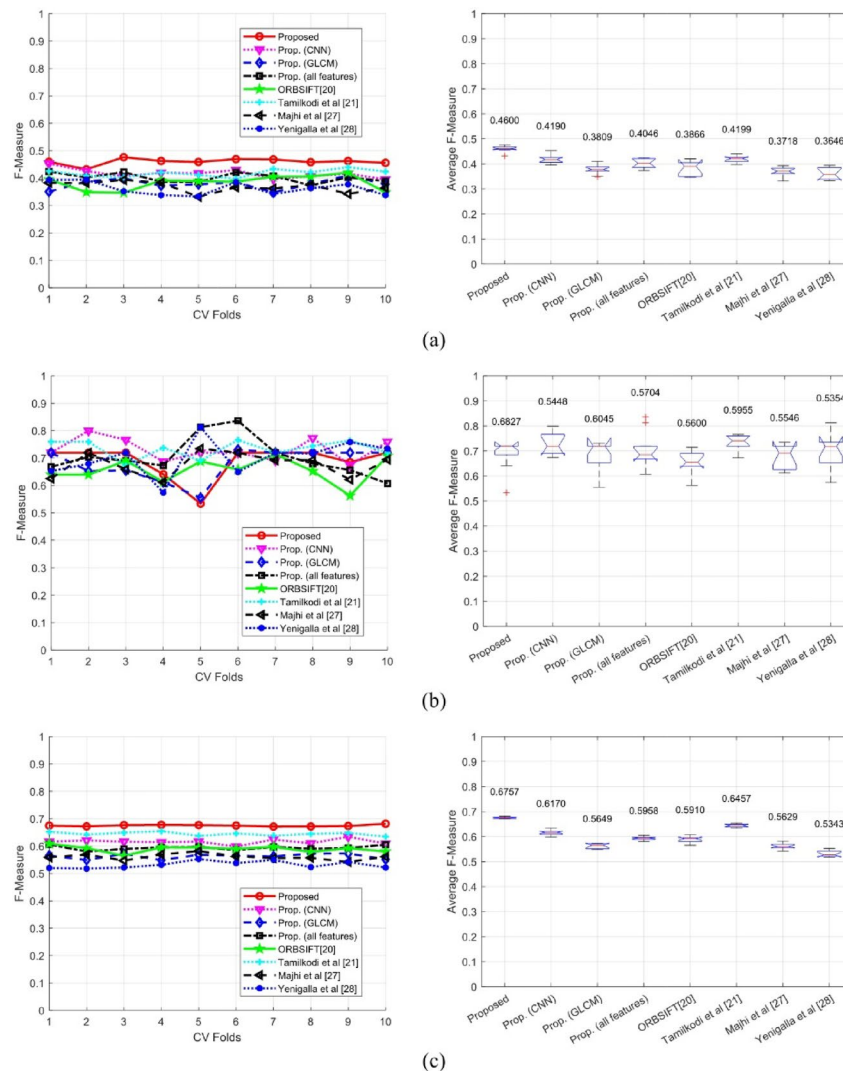


Fig. 6. Evaluation of F-Measure metric and average F-Measure (a) Corel1000, (b) Brodatz, and (c) ALOI dataset.

respectively achieved an increase of 0.05, 0.09, and 0.03, in Corel1000, Brodatz, and ALOI datasets compared to the comparative method Tamilkodi et al.²¹ Furthermore, in the box plot diagram on the right, the proposed method has shown very good performance compared to all the compared methods. The analysis of the box plot diagram allows us to identify the strengths and weaknesses of different models and select more optimal methods for image retrieval through detailed comparisons.

Figure 7 shows the Precision values separately for each class in both datasets, allowing us to examine the performance of the proposed method and other methods in image retrieval in detail. For example, in the Corel1000 dataset, our proposed method has shown the best performance in the tenth and second categories. At the same time, it has the weakest performance in the third category. In general, examining the order of performance of the proposed model in these experiments for the Corel1000 dataset shows that this method provides the best results when using all options. After that, the CNN model has also been able to show acceptable performance. However, in the case of the Brodatz dataset, the performance of the CNN model decreases significantly. The reason for this is the small number of examples in this dataset. Given that the Brodatz dataset has a small number of examples, effective training of the CNN model based on this limited data is not well possible, leading to a general decrease in the quality of its performance. Therefore, for the dataset used, the type and quality of the data strongly affects the performance of the models, and it is necessary to be careful in selecting and preparing the data. Considering the high number of categories in ALOI, this figure was not illustratable for this dataset.

Figure 8 shows the recall values of all classes in Corel1000 and Brodatz datasets. We can see that the recall values of both datasets are relatively low. That does not tell us directly that the image retrieval model performs poorly but it tells us that the number of retrievals we have performed for each graph is limited.

This limitation in the number of retrievals performed results in that we cannot extract all the relevant samples, and therefore the recall values obtained are lower than the Precision values. In other words, recall, which refers to the ability of the model to identify relevant positive examples, is significantly affected by the limited number of

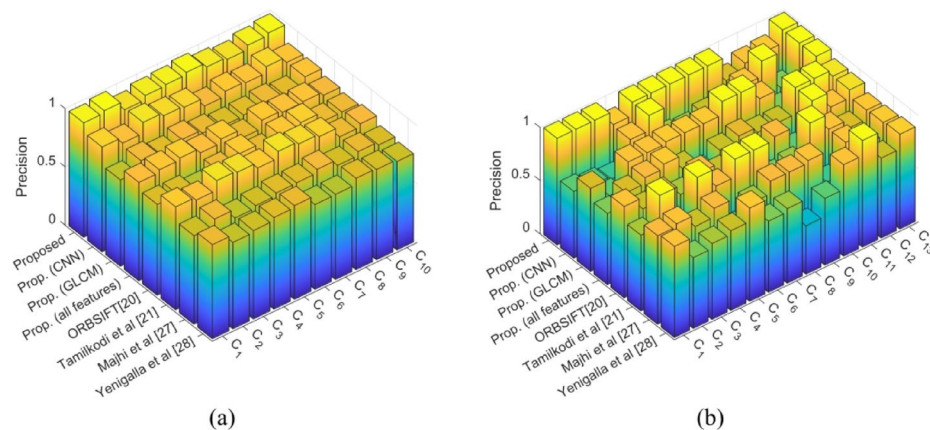


Fig. 7. Evaluation of precision metrics across different categories (a) Corel1000 and (b) Brodatz dataset.

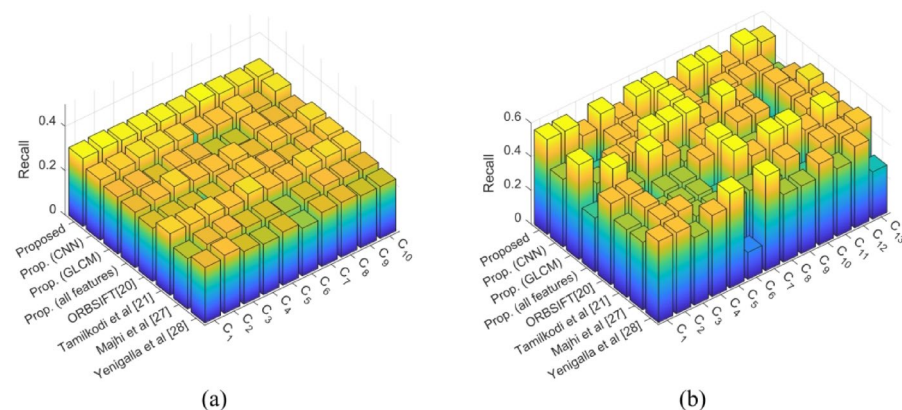


Fig. 8. Evaluation of recall metrics across different categories (a) Corel1000 and (b) Brodatz dataset.

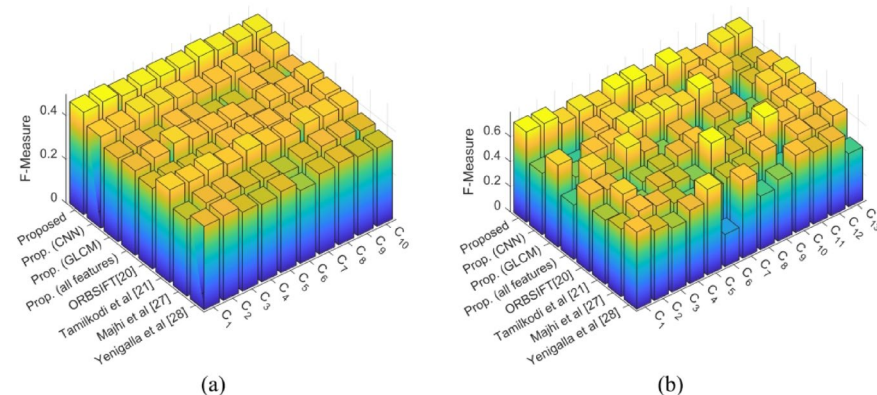


Fig. 9. Evaluation of F-Measure metrics across different categories (a) Corel1000 and (b) Brodatz dataset.

retrievals. Therefore, in the process of evaluating and analyzing the results, it should be noted that the low recall value here is due to the limited number of retrievals and not necessarily to the poor performance of the model. The same results were not presented for the ALOI dataset, because it includes a large number of categories.

Figure 9 displays the F-Measure values for all classes in the Corel1000 and Brodatz datasets. The F-Measure metric is one of the most important and widely used tools in evaluating the performance of machine learning models, especially in computational fields such as image retrieval and data classification. This metric acts as a combination of Precision and Recall and can effectively reflect the balance between these two metrics. F-Measure

allows us to have an accurate assessment of the overall performance of the model and identify its strengths and weaknesses. In this context, the F-Measure is the result of two different data sets that indicate the overall performance of our proposed method compared to other existing methods and algorithms. According to this metric, it has been observed that our proposed method has shown better performance in comparison with other methods. It infers that the proposed model can maintain the balance between the precision and recall, therefore, it can retrieve relevant images of high quality and notable accuracies. The increase in the value of F-Measure is indicative of the effective optimizations made in our proposed method and the strength of the techniques used for feature extraction, which lifted up the performance in the case of image retrieval. Considering the high number of categories in ALOI, presenting similar category-based f-measure plot was not applicable for this dataset.

Figure 10 depicts the average Precision, Recall and F-Measure values for all categories in each dataset. The results obtained from the evaluation in all iterations show that the proposed method has provided better performance in image retrieval on average. In particular, it has been able to demonstrate higher recall than other methods on the Corel1000 and Brodatz datasets. Overall, the results show that the proposed method is not only more successful in identifying related images, but also effectively increases the retrieval accuracy.

Figure 11 shows the Receiver Operating Characteristic (ROC) curves for both datasets. The ROC plot depicts the values of True Positive Rate (TPR) and False Positive Rate (FPR) and clearly indicates that our proposed method has been able to minimize FPR and, correspondingly, increase the TPR. That is, our model performed with higher accuracy in correctly identifying and retrieving target samples. In other words, the proposed method is able to effectively separate relevant images from irrelevant data by increasing the TPR, and as a result, it shows higher efficiency in image retrieval. This performance is considered a key criterion in evaluating image retrieval methods and allows us to compare the results obtained with other methods. For this reason, the ROC plot not only confirms the optimal efficiency of the proposed method, but also serves as a tool for deeper analysis of the model performance under different conditions.

Table 2 shows the results of evaluating different image retrieval methods for two datasets Corel1000 and Brodatz. In this table, the Precision, Recall, F-Measure and AUC values for each method are compared. The proposed method shows significant performance with a precision of 0.9666 and F-Measure of 0.4600 on the Corel1000 dataset, while its recall value is 0.3019, which indicates a contribution to related samples.

Concerning Brodatz set, the proposed method has the best result, a precision of 0.9527 and F-Measure of 0.6827, which shows the success of the proposed method in identifying related images. Compared with the other methods, the proposed methods like CNN and GLCM give better results but as per the proposed method, the performance is low in the aspects of Recall and F-Measure which further enhance the effectiveness of the proposed system for providing related images.

Analysis of the impact of training set size

In order to investigate the resilience and stability of our suggested model, we have performed a sensitivity analysis on the training set size. The experiment is essential to illustrate the model performance in different conditions

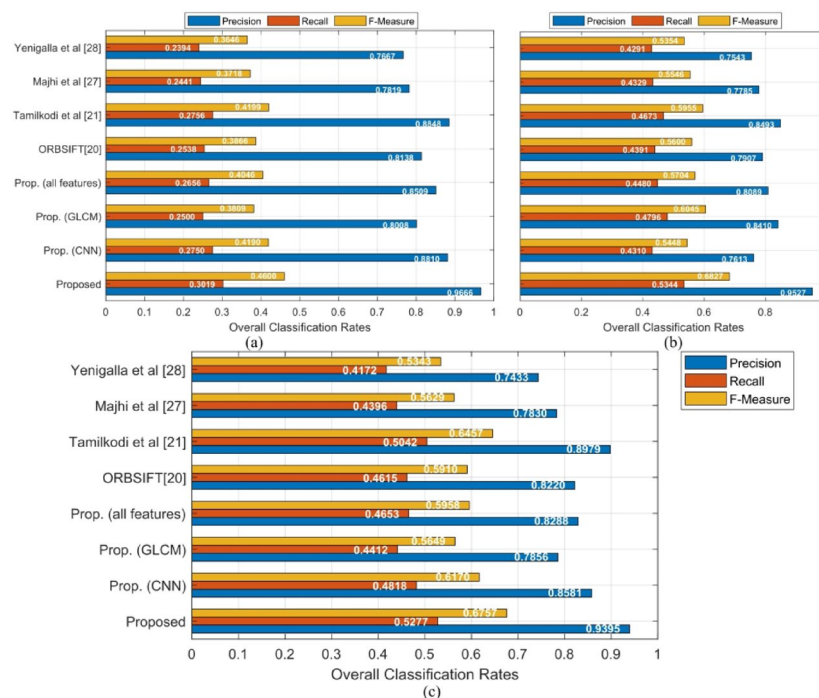


Fig. 10. Evaluation of overall Precision, Recall and F-Measure: (a) Corel1000, (b) Brodatz, and (c) ALOI dataset.

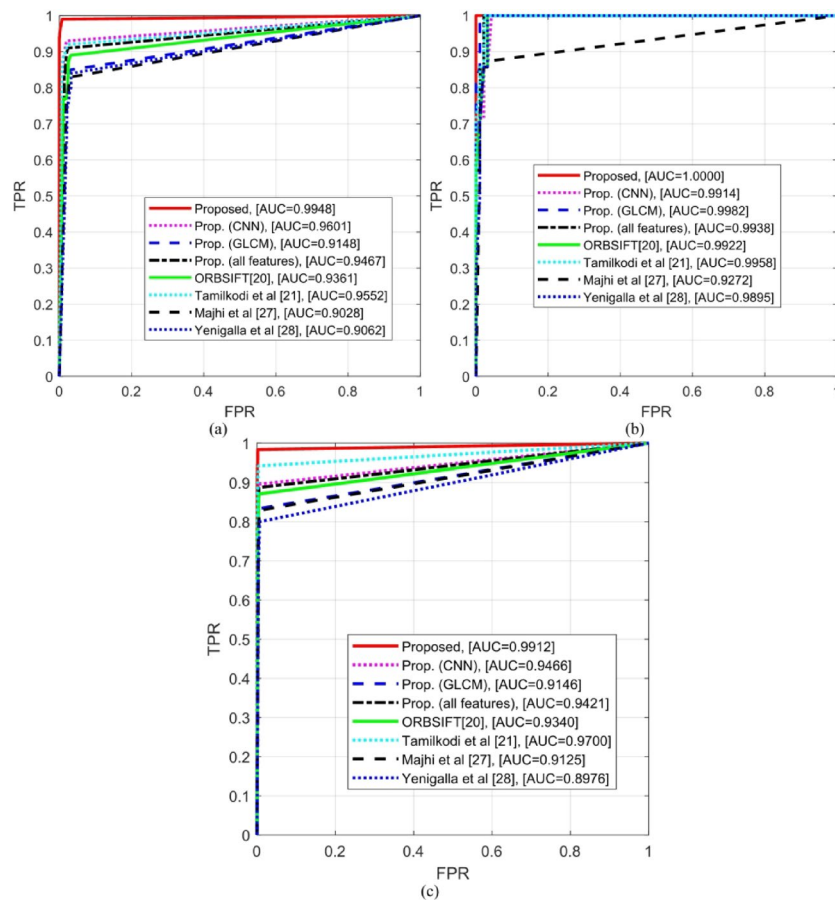


Fig. 11. evaluation of ROC curves for: (a) Corel1000, (b) Brodatz, and (c) ALOI datasets.

of the availability of data, especially where labeled data might be limited. Testing the model on smaller training sets will allow us to confirm that its good performance is not limited to the fact that it has a large training set and can perform well even when trained on a small amount of data.

To do such an analysis, we ran a different set of experiments outside the main 10-fold cross-validation. In order to emulate varying training-to-testing ratios, we divided each dataset into five different partitions of data: (10% training-90% testing), (30% training-70% testing), (50% training-50% testing), (70% training-30% testing), and (90% training-10% testing).

In order to make sure that the training and testing sets have the same distribution as the overall data, at least at lower percentages, we used stratified sampling. This method ensures that both the training and testing sets have all the categories of the images in the same ratio as they are in the original dataset. This eliminates bias in the model and gives an objective assessment in every step. Table 3 shows the findings of this analysis.

The expected and evident trend in all three datasets can be seen in Table 3: the performance metrics of the models will gracefully degrade with a decrease in the percentage of training data. This makes sense, because the model will have fewer data to learn the characteristic of each image category.

Nevertheless, the most important conclusion of this analysis is the outstanding stability of our suggested approach. The model has a high Precision and AUC in all the datasets even when trained on 10% of the data. As an example, the F-Measure is nearly 0.60 on the Brodatz and ALOI datasets even when the training set is only 10%, which means that the precision and recall are well balanced. On the same note, the AUC values are very high (over 0.94) indicating that the model has a very good ability to distinguish between various classes of images irrespective of the size of the training data.

Although the recall of the Corel1000 dataset indicates a more dramatic decrease, the precision remains over 0.83, which proves that the images retrieved are still very relevant. This analysis confirms that our feature extraction and selection pipeline generates a very discriminative feature set that enables effective learning with small data, and that the method is indeed applicable in practice where it may be difficult to obtain large labeled datasets.

Conclusion

This research identified the potential of image retrieval systems in managing images over large databases and search engines. Our proposed approach used a rich set of features in characterizing images, such as texture, color, and other patterns, against the deficiencies of the previous systems with limited feature variability. The feature

Dataset	Methods	Precision	Recall	F-Measure	AUC
Corel1000	Proposed	0.9666	0.3019	0.4600	0.9948
	Prop. (CNN)	0.8810	0.2750	0.4190	0.9601
	Prop. (GLCM)	0.8008	0.2500	0.3809	0.9148
	Prop. (all features)	0.8509	0.2656	0.4046	0.9467
	ORBSIFT [20]	0.8138	0.2538	0.3866	0.9361
	Tamilkodi et al. [21]	0.8848	0.2756	0.4199	0.9552
	Majhi et al. [27]	0.7819	0.2441	0.3718	0.9028
	Yenigalla et al. [28]	0.7667	0.2394	0.3646	0.9062
	Sezavar et al. [29]	0.8695	0.2716	0.4137	0.9581
Brodatz	Proposed	0.9527	0.5344	0.6827	0.9999
	Prop. (CNN)	0.7613	0.4310	0.5448	0.9914
	Prop. (GLCM)	0.8410	0.4796	0.6045	0.9982
	Prop. (all features)	0.8089	0.4480	0.5704	0.9938
	ORBSIFT [20]	0.7907	0.4391	0.5600	0.9922
	Tamilkodi et al. [21]	0.8493	0.4673	0.5955	0.9958
	Majhi et al. [27]	0.7785	0.4329	0.5546	0.9272
	Yenigalla et al. [28]	0.7543	0.4291	0.5354	0.9895
	Sezavar et al. [29]	0.8315	0.4646	0.5936	0.9952
ALOI	Proposed	0.9395	0.5277	0.6757	0.9912
	Prop. (CNN)	0.8581	0.4818	0.6170	0.9466
	Prop. (GLCM)	0.7856	0.4412	0.5649	0.9146
	Prop. (all features)	0.8288	0.4653	0.5958	0.9421
	ORBSIFT [20]	0.8220	0.4615	0.5910	0.9340
	Tamilkodi et al. [21]	0.8979	0.5042	0.6457	0.9700
	Majhi et al. [27]	0.7830	0.4396	0.5629	0.9125
	Yenigalla et al. [28]	0.7433	0.4172	0.5343	0.8976
	Sezavar et al. [29]	0.8860	0.4976	0.6371	0.9616

Table 2. Results of the experiments obtained for Corel1000, Brodatz, and ALOI datasets.

Dataset	Training %	Precision	Recall	F-Measure	AUC
Corel1000	10%	0.8319	0.2600	0.3961	0.9419
	30%	0.8949	0.2794	0.4257	0.9719
	50%	0.9318	0.2909	0.4433	0.9783
	70%	0.9517	0.2972	0.4528	0.9945
	90%	0.9666	0.3019	0.4600	0.9948
Brodatz	10%	0.8632	0.4788	0.6123	0.9962
	30%	0.9026	0.5070	0.6455	0.9986
	50%	0.9318	0.5186	0.6646	0.9992
	70%	0.9471	0.5282	0.6757	0.9997
	90%	0.9527	0.5344	0.6827	0.9999
ALOI	10%	0.8410	0.4723	0.6048	0.9464
	30%	0.8882	0.4988	0.6388	0.9657
	50%	0.9158	0.5143	0.6586	0.9744
	70%	0.9302	0.5224	0.6689	0.9849
	90%	0.9395	0.5277	0.6757	0.9912

Table 3. The results of the proposed model at different percentages of training data.

selection algorithm based on BHO optimized the retrieval performance and increased the processing speed, hence making it more efficient. The use of fuzzy clustering models further refined the image retrieval process based on the selected features for a more precise match. Very impressive precision results, such as 0.97, 0.95, and 0.94 on Corel1000, Brodatz, and ALOI databases respectively, have validated the efficacy of the proposed approach. This work contributed much to image retrieval and showed how feature extraction, coupled with selection, could lead to significant advances in the precision and speed of image searches.

Data availability

All data generated or analysed during this study are included in this published article.

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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